

ARUN KUMAR A R

Software Engineer | Mobile & VR Game Development
arunakar47@gmail.com | +91-7034677372 | Kollam, Kerala, India | [Portfolio](#)

PROFESSIONAL SUMMARY

Unity developer with 5+ years shipping production games and VR experiences across mobile, PC, and Oculus Quest. I specialise in gameplay systems, performance optimization, and cross-platform builds — with hands-on delivery for clients like Honeywell, KidZania, and MetaGP. Comfortable owning features end to end, mentoring junior engineers, and working closely with design, art, and QA to get things across the line.

PROFESSIONAL EXPERIENCE

Unity Developer — Game Development & VR/XR

Feb 2022 – Present

Tiltlabs Consultancy Services, Technopark Phase 1, Trivandrum, Kerala

- Owned and delivered core gameplay mechanics, feature systems, and UI pipelines across multiple shipped commercial titles — C# in Unity 2021/2022 LTS, end to end, no handholding.
- Conquered a brutal performance ceiling on Meta Quest 2, sustaining 72+ FPS by aggressively slashing draw calls, restructuring LOD systems, and rewriting URP shaders where the profiler pointed.
- Designed and built the entire game architecture for the KidZania Mahindra kiosk — draggable building systems, dynamic scoring, certificate generation, GPU-optimized lighting — a system that had to perform flawlessly for hundreds of walk-up strangers a day.
- Launched MetaGP Garage VR from scratch: first-person navigation, JWT authentication, Addressable-powered dynamic asset loading, and a cross-platform pipeline spanning Android, iOS, PC, and Oculus Quest.
- Constructed KidZania TVS Bike Studio's full live infrastructure — drag-and-drop customization, automated video capture, QR generation, AWS S3 integration — driving 100+ daily sessions with zero critical downtime.
- Pioneered the networking layer for GenRobotics VR: real-time TCP socket video streaming on Oculus Quest, designed from a blank page because nothing off the shelf came close to what the project demanded.
- Diagnosed and dismantled the worst performance bottlenecks in Honeywell's industrial VR training app — hunted them through the Unity Profiler methodically and unlocked a 40%+ improvement in training throughput.
- Champion code reviews and have mentored several junior engineers — my approach is to explain the why behind every decision, because that's what stops the same bug from appearing in three future PRs.
- Harness GitHub Copilot and Cursor daily to accelerate prototyping and decompose complex features — but I know exactly when to put the tools down and think it through myself.
- Collaborate tightly with designers, artists, and QA — I've found that clear, direct cross-discipline communication is what separates games that ship from games that stall.

Unity Developer — 3D, VR & Game Development

Mar 2021 – Feb 2022

Embright Infotech Pvt Ltd, Technopark Phase 1, Trivandrum, Kerala

- Shipped a VR Laser and Light Reflection Simulation on Oculus Quest — accurately modelling laser physics across multiple material surfaces forced me to master Unity Physics at a level most mobile projects never demand.
- Crafted an Alphabet Learning mobile app for Android/iOS: drag-and-drop gameplay, 3D touch-rotation interactions, and a forgiving UX built with young users at the centre of every design decision.
- Contributed to Auticare VR, an ASD educational platform — drove feature design, implementation, and QA for interactive modules that had to serve a sensitive, specific user group without a single rough edge.
- Contributed to ULCC Archviz through meticulous baked lighting optimizations, transforming heavy scenes into smooth, photorealistic real-time walkthroughs.

TECHNICAL SKILLS

Languages:	C# (expert — daily driver for 5+ years), C++ (C++14/17, STL — actively mastering) and C
Game Engines:	Unity 3D 2019/2021/2022 LTS — deeply fluent. Unreal Engine 5 exposure; strong grasp of engine architecture that transfers across platforms.
Mobile/Platforms:	Android, iOS, PC, WebGL, Oculus/Meta Quest 2 — battle-tested cross-platform pipelines, not theoretical ones.
Graphics:	Unity URP, Shader Graph, LOD, draw call reduction, baked/dynamic lighting, post-processing, OpenGL ES concepts
Architecture:	OOP, SOLID, MVC, Addressable Asset System, JSON serialization — I build systems that the next engineer can actually navigate.
Networking:	TCP/IP sockets, REST APIs, JWT auth, AWS S3. Expanding into Photon, Netick 2 for multiplayer.
Performance:	Unity Profiler, memory management, frame rate optimization, scene & asset streaming — this is the problem I chase hardest.
Tools:	Visual Studio, Rider, Git, Xcode
AI Tools:	GitHub Copilot, Cursor — leverage them to move faster, not to think less.

KEY PRODUCTION PROJECTS

- MetaGP Garage VR — Engineered a first-person VR experience with JWT auth and Addressable dynamic loading, sustaining 72+ FPS on Meta Quest 2 across Android and PC simultaneously.
- Honeywell VR Training — Transformed a sluggish industrial petroleum plant simulation into a high-fidelity training tool; unlocked 40%+ efficiency gains through relentless profiling and targeted rebuilds.
- KidZania TVS Bike Studio — Powered 100+ live daily sessions with a custom asset pipeline, QR generation, and AWS S3 integration, built entirely from scratch.
- GenRobotics VR — Pioneered real-time TCP socket video streaming on Oculus Quest; invented the protocol because no existing solution was adequate.
- Serenity PathWays — Delivered a clinical cognitive training VR app with physics-driven puzzles, progressive difficulty curves, and spatial audio tuned for genuine therapeutic impact.
- Auticare VR — Shaped an ASD educational platform's core interactive modules, balancing technical precision with deep sensitivity to the end user's needs.

EDUCATION

Bachelor of Technology — Computer Science & Engineering
College of Engineering, Kottarakara

2016 – 2020 | CGPA:
7.4/10